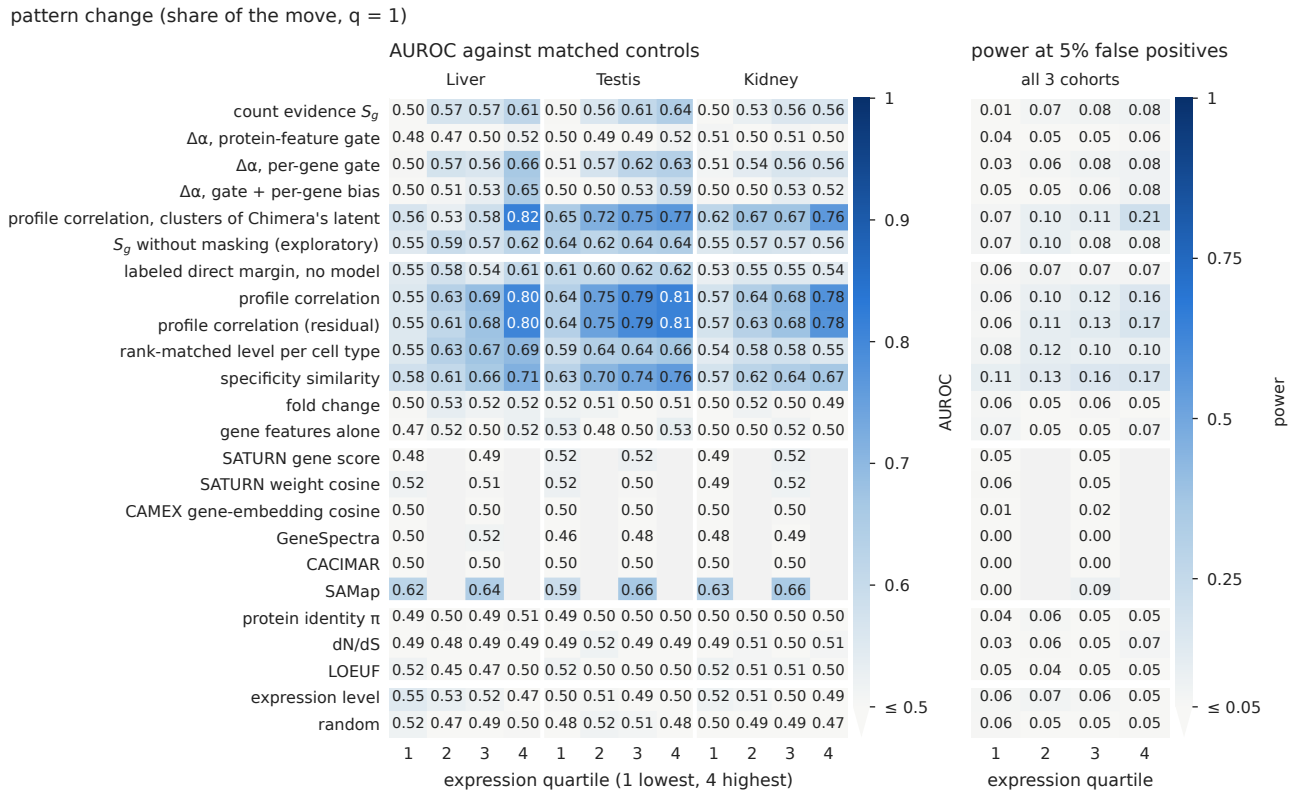
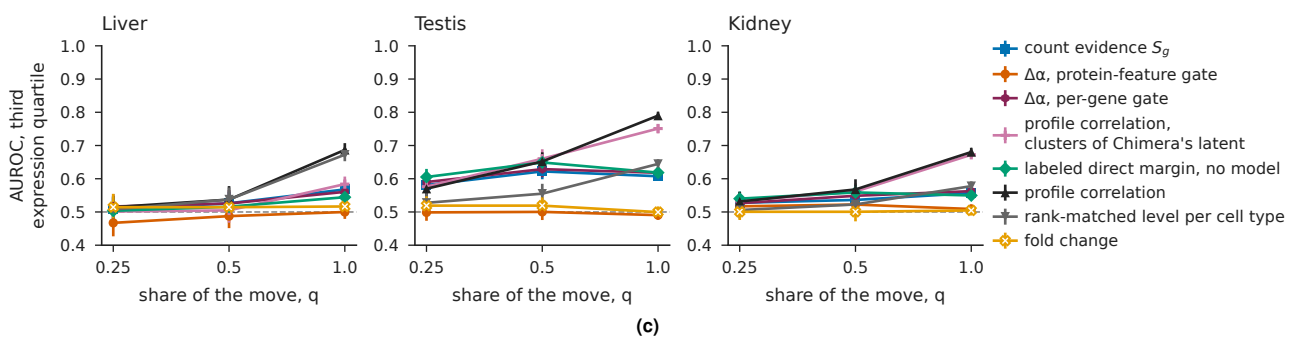
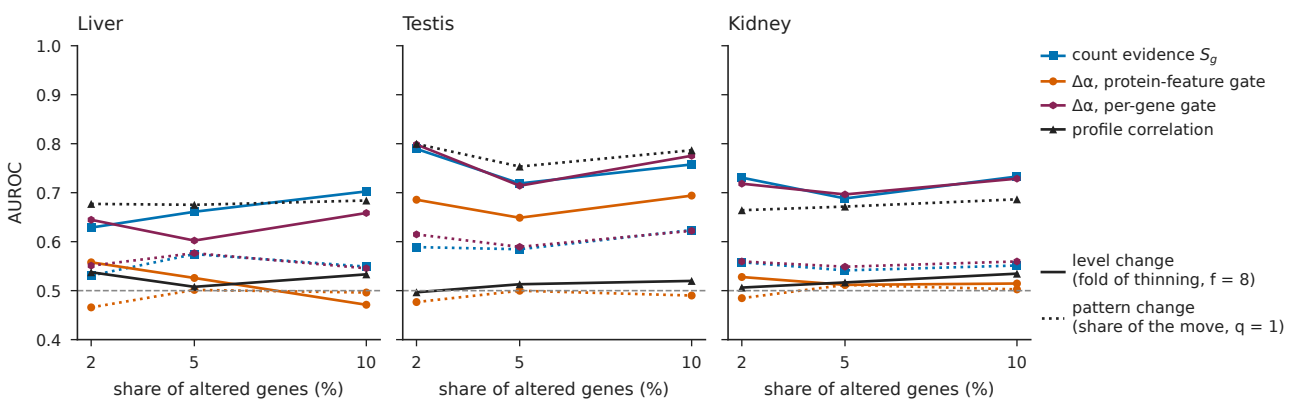

pattern change (share of the move, q = 1)
AUROC against matched controls
power at 5% false positives
Liver
Testis
Kidney
all 3 cohorts
count evidence S_g
Δα, protein-feature gate
Δα, per-gene gate
Δα, gate + per-gene bias
profile correlation, clusters of Chimera's latent
S_g without masking (exploratory)
labeled direct margin, no model
profile correlation
profile correlation (residual)
rank-matched level per cell type
specificity similarity
fold change
gene features alone
SATURN gene score
SATURN weight cosine
CAMEX gene-embedding cosine
GeneSpectra
CACIMAR
SAMap
protein identity π
dN/dS
LOEUF
expression level
random
AUROC
power
expression quartile (1 lowest, 4 highest)
expression quartile


Liver
Testis
Kidney
AUROC, third expression quartile
share of the move, q
count evidence S_g
Δα, protein-feature gate
Δα, per-gene gate
profile correlation, clusters of Chimera's latent
labeled direct margin, no model
profile correlation
rank-matched level per cell type
fold change

(c)


Liver
Testis
Kidney
AUROC
share of altered genes (%)
count evidence S_g
Δα, protein-feature gate
Δα, per-gene gate
profile correlation
level change (fold of thinning, f = 8)
pattern change (share of the move, q = 1)

(d)